MSG No. 63.461 $Cmcm'$ [Type III, orthorhombic]

Table 1: Wyckoff site: `4a`, site symmetry: `2'/m..`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,3,5,7]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[2,4,6,8]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[9,11,13,15]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[10,12,14,16]`

Table 2: Wyckoff site: `4b`, site symmetry: `2'/m..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,3,5,7]`
2	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,4,6,8]`
3	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[9,11,13,15]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[10,12,14,16]`

Table 3: Wyckoff site: `4c`, site symmetry: `m2'm'`

No.	position	mapping
1	$\left[0,\ y,\ \frac{1}{4}\right]$	`[1,3,6,8]`
2	$\left[0,\ -y,\ \frac{3}{4}\right]$	`[2,4,5,7]`
3	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[9,11,14,16]`
4	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[10,12,13,15]`

Table 4: Wyckoff site: `8d`, site symmetry: `-1'`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ 0\right]$	`[1,15]`
2	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}\right]$	`[2,16]`
3	$\left[\frac{3}{4},\ \frac{1}{4},\ 0\right]$	`[3,13]`
4	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}\right]$	`[4,14]`
5	$\left[\frac{1}{4},\ \frac{3}{4},\ 0\right]$	`[5,11]`
6	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}\right]$	`[6,12]`
7	$\left[\frac{3}{4},\ \frac{3}{4},\ 0\right]$	`[7,9]`
8	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}\right]$	`[8,10]`

Table 5: Wyckoff site: `8e`, site symmetry: `2'..`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,5]`
2	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[2,6]`
3	$[-x,\ 0,\ 0]$	`[3,7]`
4	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[4,8]`

continued ...

Table 5

No.	position	mapping
5	$\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[9,13]`
6	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[10,14]`
7	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$	`[11,15]`
8	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[12,16]`

Table 6: Wyckoff site: `8f`, site symmetry: `m..`

No.	position	mapping
1	$[0,\ y,\ z]$	`[1,3]`
2	$\left[0,\ -y,\ z+\frac{1}{2}\right]$	`[2,4]`
3	$[0,\ -y,\ -z]$	`[5,7]`
4	$\left[0,\ y,\ \frac{1}{2}-z\right]$	`[6,8]`
5	$\left[\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	`[9,11]`
6	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[10,12]`
7	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	`[13,15]`
8	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[14,16]`

Table 7: Wyckoff site: `8g`, site symmetry: `..m'`

No.	position	mapping
1	$\left[x,\ y,\ \frac{1}{4}\right]$	`[1,8]`
2	$\left[-x,\ -y,\ \frac{3}{4}\right]$	`[2,7]`
3	$\left[-x,\ y,\ \frac{1}{4}\right]$	`[3,6]`
4	$\left[x,\ -y,\ \frac{3}{4}\right]$	`[4,5]`
5	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[9,16]`
6	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[10,15]`
7	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[11,14]`
8	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[12,13]`

Table 8: Wyckoff site: `16h`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	`[2]`
3	$[-x,\ y,\ z]$	`[3]`
4	$\left[x,\ -y,\ z+\frac{1}{2}\right]$	`[4]`
5	$[x,\ -y,\ -z]$	`[5]`
6	$\left[-x,\ y,\ \frac{1}{2}-z\right]$	`[6]`
7	$[-x,\ -y,\ -z]$	`[7]`
8	$\left[x,\ y,\ \frac{1}{2}-z\right]$	`[8]`
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	`[9]`
10	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[10]`
11	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z\right]$	`[11]`

continued ...

Table 8

No.	position	mapping
12	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[12]
13	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[13]
14	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[14]
15	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ -z\right]$	[15]
16	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[16]